MagicCode Super-Resolution Software Error Code Manual

Effective Date: December 1, 2025

Applicable Version: All versions of MagicCode Super-Resolution Software

1. Overview

This manual defines the error codes generated by the MagicCode Super-Resolution Software (hereinafter referred to as "Software") during runtime, including their detailed descriptions, trigger scenarios, and basic troubleshooting directions. It serves as a reference for developers, technical support personnel, and end users to quickly identify and resolve issues related to the Software.

The error codes of the Software are structured with hexadecimal notation, where the high-order bits indicate the error category (e.g., model parameter errors, memory errors) and the low-order bits indicate specific error types. The core error categories covered in this manual include **Model Parameter Errors**, **Memory Management Errors**, **Pointer & Handle Errors**, and **Control Command Errors**.

2. Error Code Details

2.1 Model Parameter & File Errors

This category of errors (error code prefix: 0x100000xx) is triggered during the loading, parsing, or validation of super-resolution model parameters and model files, which are core prerequisites for the Software's operation.

Error Code	Error Description	Trigger Scenarios & Troubleshooting
0x10000008	Invalid model parameter; not "veryfast" or "fast"	Triggered when the super-resolution mode parameter passed to the Software is not one of the supported values ("veryfast" or "fast"). Troubleshooting: Verify that the mode parameter

		is set to the correct enumerated value as specified in the Software API documentation.
0x10000009	Super-resolution scaling factor in model parameters exceeds 4x	Triggered when the set super-resolution scaling ratio (e.g., 5x, 6x) exceeds the Software's maximum supported ratio (4x). Troubleshooting: Adjust the scaling factor to one of the supported integer multiples (2x, 3x, 4x) as defined in the Software Specification.
0x1000000A	Failed to open the model parameter file	Triggered when the Software cannot access the specified model file (e.g., file path is incorrect, file is missing, or insufficient read permissions). Troubleshooting: Check the model file path for accuracy, ensure the file exists, and verify that the running account has read permissions for the file.
0x1000000B	Failed to read the length of the model file	Triggered when the Software fails to retrieve the size information of the model file (e.g., file is corrupted, storage device is faulty). Troubleshooting: Check if the model file is intact, run a storage device health check, and re-download or replace the model file if necessary.

0x1000000C	Abnormal length of the model file	Triggered when the actual length of the model file does not match the preset standard length (e.g., file is incomplete due to interrupted download). Troubleshooting: Re-download the model file from the official channel and verify its integrity using the provided MD5/SHA256 checksum.
0x1000000D	Incorrect model file type	Triggered when the file specified as the model is not in the supported format (e.g., using a text file instead of a binary model file). Troubleshooting: Ensure the model file is in the format specified by the Software (e.g., .mcmod) and obtained from the official source.
0x1000000E	Error in Model File Parameter 1	Triggered when the first core parameter in the model file (e.g., input resolution constraint) is invalid or out of range. Troubleshooting: Replace the model file with a valid one from the official channel; contact technical support if the issue persists.
0x1000000F	Error in Model File Parameter 2	Triggered when the second core parameter in the model file (e.g., algorithm kernel version) is invalid or incompatible. Troubleshooting: Ensure

		the model file version matches the Software version; update either the Software or the model file to a compatible version.
0x10000010	Error in Model File Parameter 3	Triggered when the third core parameter in the model file (e.g., SIMD instruction set compatibility flag) is invalid. Troubleshooting: Use a model file compatible with the device's CPU architecture (x86_64/ARM64) and its supported SIMD instructions (AVX2/NEON).
0x10000011	Abnormal reading of model file length	Triggered when the model file length read by the Software is inconsistent with the actual file size (e.g., file system error, disk fragmentation). Troubleshooting: Run a file system repair tool (e.g., chkdsk for Windows, fsck for Linux) and recheck the model file.
0x10000012	Error in Model File Parameter 4	Triggered when the fourth core parameter in the model file (e.g., output pixel format) is invalid. Troubleshooting: Replace the model file with an official version; avoid modifying the model file's internal parameters manually.

0x10000013	Conflict between set VERYFAST_SR mode and model file parameters	Triggered when the model file is not compatible with the selected "veryfast" super-resolution mode (e.g., model is optimized for "fast" mode only). Troubleshooting: Use a model file marked as compatible with "veryfast" mode or switch to "fast" mode.
0x10000014	Conflict between set FAST_SR mode and model file parameters	Triggered when the model file is not compatible with the selected "fast" super-resolution mode (e.g., model is designed for "veryfast" mode only). Troubleshooting: Use a model file marked as compatible with "fast" mode or switch to "veryfast" mode.
0x10000016	Error in Model File Parameter 5	Triggered when the fifth core parameter in the model file (e.g., multi-threading control flag) is invalid. Troubleshooting: Re-download the model file from the official website; ensure the Software is updated to the latest version.
0x1000001A	CPU instruction set not supported	Triggered when the device's CPU does not support the required SIMD instruction sets (AVX2 for x86_64, NEON for ARM64). Troubleshooting: Check if the device meets the

		Software's hardware requirements; upgrade the hardware or use a device with a compatible CPU.

2.2 Memory Management Errors

This category of errors (error code prefix: 0x200000xx, 0xA00000xx) is related to memory allocation, release, and leak issues during the Software's operation, which directly affect the stability of the program.

Error Code	Error Description	Trigger Scenarios & Troubleshooting
0x20000002	Failed to allocate Memory 1	Triggered when the Software fails to allocate the first block of required memory (e.g., for model loading) due to insufficient system memory or memory fragmentation. Troubleshooting: Close other memory-intensive applications, increase the system's available memory, or reduce the input image resolution.
0x20000003	Failed to allocate Memory 2	Triggered when the Software fails to allocate the second block of required memory (e.g., for intermediate image processing). Troubleshooting: Same as 0x20000002; ensure the device's physical memory meets the Software's minimum requirements (refer to the Software Specification).

0x20200001	Memory allocation failed	A general memory allocation error triggered when the Software cannot obtain sufficient memory for any runtime operation. Troubleshooting: Restart the Software and close background applications; for persistent issues, upgrade the device's memory or use lower-resolution input content.
0x10300001	Abnormal memory release	Triggered when the Software fails to release allocated memory properly (e.g., double free, release of unallocated memory). Troubleshooting: Update the Software to the latest version to fix known memory management bugs; avoid using the Software with modified or pirated code.
0xA0000001	Suspected memory leak	Triggered when the Software's memory usage continues to increase without release during long-term operation. Troubleshooting: Restart the Software periodically; update to the latest version which may include memory leak fixes; report the issue to technical support with runtime logs.

2.3 Pointer & Handle Errors

This category of errors (error code prefix: 0x101000xx) involves invalid pointers or abnormal handle operations, which are often related to incorrect API calls or internal program exceptions.

Error Code	Error Description	Trigger Scenarios & Troubleshooting
0x10100001	Handle space is abnormally overwritten	Triggered when the Software's core handle (e.g., model handle, device handle) is corrupted by invalid memory operations (e.g., buffer overflow). Troubleshooting: Restart the Software; ensure no third-party software is interfering with the Software's memory space; update the Software to fix potential security vulnerabilities.
0x10100002	Null input image pointer when calling MC_Process	Triggered when the input image pointer passed to the MC_Process API is NULL (e.g., incorrect parameter passing in user-developed code). Troubleshooting: Verify that the input image data is properly initialized and the pointer is valid before calling MC_Process; refer to the API Documentation for correct parameter usage.
0x10100003	Abnormal internal pointer in the algorithm	Triggered by internal program exceptions (e.g., uninitialized pointers in the super-resolution algorithm). Troubleshooting: Update the Software to the latest

		stable version; if the issue persists, collect runtime logs and contact technical support.

2.4 Control Command Errors

This category of errors (error code prefix: 0x102000xx) occurs during the execution of control commands via the MC_Control API, typically due to invalid parameters or incorrect command usage.

Error Code	Error Description	Trigger Scenarios & Troubleshooting
0x10200001	Null output parameter pointer when calling MC_Control with QUERY_STATUS	Triggered when the output parameter pointer for status query is NULL when calling MC_Control with the QUERY_STATUS command. Troubleshooting: Initialize the output parameter structure and pass a valid pointer before calling the command; refer to the API Documentation for command syntax.
0x10200003	Abnormal cmd parameter in MC_Control	Triggered when the cmd parameter passed to the MC_Control API is not a supported command (e.g., invalid command code). Troubleshooting: Verify that the cmd parameter is one of the enumerated values specified in the API Documentation; update the API header file to ensure command code consistency.

3. Error Handling Guidelines

3.1 General Troubleshooting Steps

1. **Verify Basic Conditions**: Ensure the Software version is compatible with the device's OS and CPU architecture; confirm the model file is valid and from the official channel.
2. **Check Parameters**: Review the super-resolution mode, scaling factor, and API parameters to ensure they conform to the Software's specifications.
3. **Optimize System Resources**: Close unnecessary background applications to free up memory; ensure the storage device has sufficient space and normal read/write capabilities.
4. **Update Software & Model Files**: Install the latest version of the Software and corresponding model files to fix known bugs.
5. **Collect Debug Information**: If the issue persists, collect the error log (generated automatically by the Software in the "logs" directory), device hardware information, and operation steps, then submit them to technical support.

3.2 Contact Technical Support

For errors that cannot be resolved with the above steps, contact MagicCode's technical support team via the following channels:

- Technical Support Email: contact@magiccode.com

When submitting a support request, please include the error code, error log, Software version, device model, and detailed operation steps to facilitate efficient problem resolution.

4. Revision History

Version	Date	Revision Content
V1.0	2025-12-01	Initial release, covering all specified error codes and troubleshooting guidelines